

Project Overview:

For our project, we are aiming to utilize the Gemma models provided to determine how well they could provide educational and informative answers to different science questions along different topics and educational levels. The dataset that we used comes from [Kaggle](#), and consists of 50,000 rows of data with the following columns: id, subject (chemistry, physics, biology), grade_level (elementary, middle, high), topic (80 different science topics), student_input (the prompt for the tutor; in our case, the LLM), tutor_response (an response provided by a knowledgeable tutor providing age-appropriate explanations), data_type (conversation, homework_help, direct_qa), and intent (educational_help, homework_assistance, knowledge_query). Our goal is to measure whether the model can provide accurate, adaptive, and relevant information in an efficient manner and with efficient usage of resources. For our project, we hope to measure and evaluate the following: **accuracy** (how accurate in terms of *cosine similarity* is the model to real answers provided by tutors), **adaptability** (how adaptive is the model to the depth of a response, such as accounting for student level and topic), and **stubbornness** (how much the model changes its responses). The key metric for our project is our **adaptability** metric, targeting how our model can adapt and cater to the needs of a student based on the subject and grade level information. To measure **adaptability**, we are going to prompt the model with the following information in three different intervals: the individual student_input, the student_input and subject, and lastly the student_input, subject, and grade_level, investigating how our model performs as a baseline and with additional context. In addition to **adaptability**, we plan on expanding our analysis with **stubbornness**, which measures how our model responses change at each pass. To measure our model over different languages, we'll be testing performance with *Spanish* to see whether the model performs differently. Can our model

adapt well to help provide accurate, adaptive, and reliable tutor answers in an efficient and effective way to help educate students?

Code on GitHub: <https://github.com/EllingsenMP/6051HackathonBestTeam/tree/main>

Table of Results:

Metric	Result	Why it needs to be included	How it's measured (automated, LLM-as-judge, or human-required)	Limitations
Accuracy	0.26 (English), 0.29 (Spanish)	We need our model to accurately provide answers to better educate students. Otherwise, irrelevant and harmful teaching would be provided to students hoping to learn.	Cosine similarity to our prelabeled responses in the dataset	Does not account for the subject matter or student level
Adaptability	0.26, 0.23, 0.27 (English), 0.29, 0.30, 0.29 (Spanish)	Can the model adapt the depth of response as the subject and grade level of the student changes?	Tiered cosine similarity and increase in cosine similarity at each level	Can be computationally expensive
Stubbornness	0.60, 0.53 (English), 0.66, 0.64 (Spanish)	How stubborn is our model after making each	Cosine similarity between	Scale and potentially limited detail/information

		pass?	responses after each pass	
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Written Discussion:

In order to test all of our evaluation metrics of accuracy, adaptability, and latency and cost, we first prompted the Gemma LLM to answer 30 of our dataset's elementary science questions in batches and 3 passes (first pass: just prompting with the original question; second pass: including subject for each question; third pass: including subject and user's grade level for each question). Additionally, to test the model's cross-linguistic abilities, we translated the same 30 questions into Spanish and repeated the above prompting process. While we did try to replicate this with Bulgarian as well, we ran into character issues while translating and trying to run our code on that dataset that ended up corrupting the file. Once we had two different datasets for each language, we were able to judge accuracy by testing against the preliminary datasets' prompt responses, adaptability through tiered cosine similarity between preliminary data and Gemma-generated responses for each of the passes (defined above), stubbornness through cosine similarity between only the Gemma-generated responses across grade levels to test for changes in complexity of language (e.g., Gemma should not respond to an elementary schooler in the same way it responds to a high schooler), and latency and cost through displayed run time.

Given the results of our model, it appears that there are different concerns present with how our Gemma model performs based on **accuracy**, along with our **adaptability** and **stubbornness** metrics. First, our model was not extremely similar to the educator/tutor response, with accuracy scores under 0.3. While it's not a labeled response as accuracy measures are typically used, it does indicate that the response provided by the Gemma model deviated a good amount from the educator response. Furthermore, the Gemma model was not adaptive

throughout each pass, and providing additional context like subject or grade level did not do all that much to contribute to the model's cosine similarity to the educator response. Lastly, in terms of stubbornness scores, the model shared similar cosine similarities indicating that there's little prompting that will influence the responses.

Since we evaluated the Gemma responses in both English and Spanish, we were able to compare across the two languages. There is a noticeable difference in average scores for specifically our stubbornness metric; on average, the Gemma responses did not change much when more context was added in the prompt passes. This indicates that Gemma is not able to adapt as well in languages other than English. While we were not able to test the model with the data in Bulgarian, we expect that Gemma would've performed even worse because this is a low-resource language.

What each individual contributed:

Marissa Burton:

- Found and prepared data for domain
- Wrote written discussion
- Contributed to evaluation metrics
- Contributed to video

Max Ellingsen:

- Translator of languages
- Contributed to evaluation metrics
- Github organizer
- Edited LLM judge script
- Contributed to video

Jarrett Markman:

- Writer of project overview
- Contributed to evaluation metrics
- Edited LLM judge script
- Contributed to video

Cole Whittington:

- Main driver of code
- Contributed to evaluation metrics
- Contributed to video